

SaaS Customer Churn Analysis

Retention Risk & Revenue Impact-Executive Summary

Analyst: Siddhi Goswami | Dataset: IBM Telco Customer Churn (7,043 customers) | Tools: SQL, Tableau

BUSINESS PROBLEM

A B2B SaaS telecommunications company is losing **\$1,669,570 annually** to preventable customer churn. At a **26.54% overall churn rate** — nearly 4x the SaaS industry benchmark of 5-7% — the company churns roughly 1 in 4 customers per year. The average churned customer was paying \$74.44 per month, representing \$893 in lost annual recurring revenue per customer. This analysis identifies the root causes of churn, validates a risk scoring model across the full customer base, and delivers prioritized retention recommendations with projected revenue recovery.

KEY FINDINGS

- **CONTRACT TYPE IS THE #1 CHURN DRIVER:** Month-to-month customers churn at 42.71% — 15x the rate of two-year contract customers at 2.83%. The 3,875 customers on month-to-month contracts represent the single largest addressable churn risk.
- **NEW CUSTOMERS ARE THE HIGHEST-RISK GROUP:** 52.94% of customers churn within their first 6 months — more than 1 in 2 new customers leave before completing half a year. A LAG window function trend analysis confirmed the steepest improvement occurs between months 0-6 and 7-12 (a 17.05 point drop), identifying the first 6 months as the critical retention window.
- **FIBER OPTIC SERVICE HAS A SYSTEMIC PROBLEM:** Premium fiber optic customers churn at 41.89% — double the DSL rate of 18.96% — despite paying the highest prices. All top 5 highest-churn customer profiles involve fiber optic, indicating an expectation-delivery gap at the product level.
- **PAYMENT METHOD SIGNALS DISENGAGEMENT:** Electronic check customers churn at 45.29% — 3x the rate of automatic payment customers (15-17%). Manual payment behavior is a reliable proxy for low customer commitment.
- **HIGH-VALUE CUSTOMERS CARRY 81% OF REVENUE RISK:** Customers paying \$70+ per month account for \$1,355,772 — 81% — of total annual churn revenue loss, despite representing 51% of the customer base.
- **RISK MODEL VALIDATED:** A multi-factor churn risk scoring model (built using chained CTEs) confirmed High Risk customers churn at 49.64% — nearly 10x the 5.1% Low Risk rate. 1,384 active High Risk customers represent \$940,218 in revenue to protect proactively before they churn.

RECOMMENDATIONS

1. CONTRACT UPGRADE CAMPAIGN — *Highest Priority*

Target month-to-month customers in their first 90 days with a 10-15% discount to switch to an annual contract. A 10% conversion rate removes 387 customers from the highest-risk category immediately.

2. AUTO-PAY ENROLLMENT INCENTIVE

Offer a one-time \$10-20 bill credit to electronic check customers who switch to automatic payment. Targets the 2,365 highest-churn payment segment with a low-cost, high-impact intervention.

3. FIBER OPTIC ONBOARDING PROGRAM

Implement structured 30/60/90-day check-in calls and proactive support outreach for all new fiber optic customers. Addresses the expectation-delivery gap before dissatisfaction leads to cancellation.

4. HIGH-VALUE CUSTOMER RETENTION PROGRAM

Assign dedicated customer success coverage to the top 100 highest-value at-risk customers identified in the SQL priority analysis. Each customer has a specific recommended action based on their individual risk profile.

PROJECTED BUSINESS IMPACT

A combined retention program targeting all four recommendations above is projected to reduce annual churn by 20-30%:

Scenario	Churn Reduction	Annual Revenue Recovered
Conservative	20% improvement	\$333,914
Moderate	30% improvement	\$500,871

Program ROI: Estimated retention program cost of \$50,000-150,000 annually against a recovery of \$333,000-500,000 yields a return of **\$2.20-\$3.30 per dollar invested**. The program pays for itself at even the most conservative churn reduction estimate.

Full Analysis: <https://github.com/siddhigoswami003/saas-churn-analysis>

Dashboard: <https://public.tableau.com/app/profile/siddhi.goswami/viz/TelCoSaaSChurnAnalysis/ChurnAnalysisDashboard?publish=yes>